W² Education Standards Authority

Centre Number

Student Number

2077 **HIGHEST SCHOOL CERTIFICATE PREMIERE**

Mathematics Extension 2

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Write your Centre Number and Student Number at the top of this page
- This paper is for the exclusive internal use of W² Education students, and no permission or authorisation is granted for redistribution

Total marks: 100

Section I – 10 marks (pages 2–6)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 7–16)

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks
Attempt Questions 1–10
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

---

1 Which of the following is equal to $\sin\theta + i\cos\theta$?

A. $ie^{-i\theta}$

B. $ie^{i\theta}$

C. $-ie^{-i\theta}$

D. $-ie^{i\theta}$

2 Consider the statement

"Not all that glitters is gold."

Which of the following is the negation?

A. All things that glitter are NOT gold

B. All that glitters is gold

C. Something glitters that is NOT gold

D. Something that is gold does NOT glitter

3 What is the value of $\left|e^{6i} + e^{8i}\right|$?

A. $2\sin\left(\frac{1}{2}\right)$

B. $2\sin 1$

C. $2\cos\left(\frac{1}{2}\right)$

D. $2\cos 1$

4 Let n be an odd positive integer greater than 2 that is NOT prime.

Let k be a prime divisor of n.

Which of the following statements must be true?

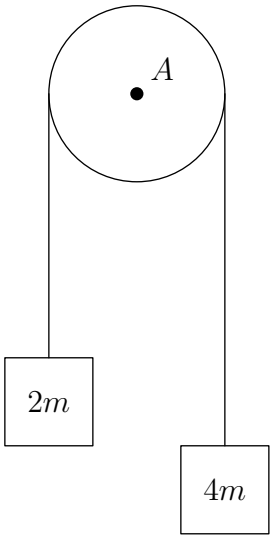
A. $k \leq \sqrt{n}$

B. $k \geq \sqrt{n}$

C. $k \leq \dfrac{n}{3}$

D. $\dfrac{n}{k}$ must be prime

5 A light string passes a smooth pulley fixed at A. Two masses, $4m$kg and $2m$kg, are attached to the ends of the string, as shown.



The acceleration due to gravity is $g$ms^{-2}.

The masses are moving vertically in a resistive medium, where the resistive force on each mass has magnitude kv, where v is the speed of the mass.

What is the terminal speed of the larger mass?

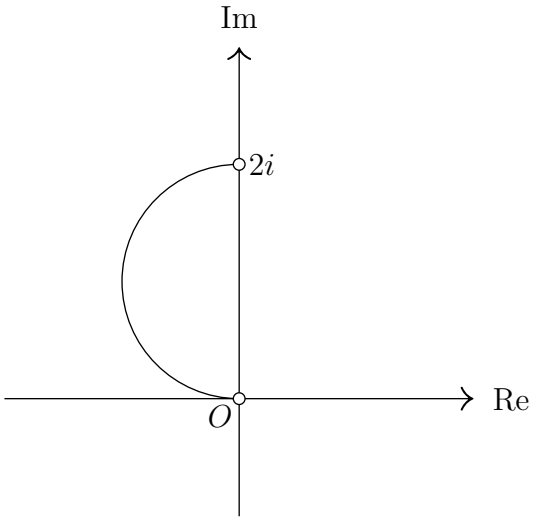
A. $\dfrac{mg}{k}$

B. $\dfrac{2mg}{k}$

C. $\dfrac{3mg}{k}$

D. $\dfrac{4mg}{k}$

6 The locus in the complex plane defined by $\text{Arg}(z - 2i) - \text{Arg}(z) = \theta$ is a semicircle with its endpoints removed, as shown.



Which of the following statements about the value of θ is true?

A. $|\theta| = \dfrac{\pi}{2}$

B. $0 < \theta < \dfrac{\pi}{2}$

C. $\dfrac{\pi}{2} < \theta < \pi$

D. $\theta < -\pi$

7 A particle is released from rest at point A and moves in simple harmonic motion. Upon reaching point B, which is 4 metres to the left of A, the particle turns back towards A, and reaches it after some time.

At the first instant the particle is 1 metre from its central point of motion, its velocity is equal to its acceleration.

How many seconds elapse between this instant and the first instant the particle is 1 metre from its central point of motion while moving in the opposite direction to its initial motion?

A. $\dfrac{\pi}{3\sqrt{3}}$

B. $\dfrac{\pi}{\sqrt{3}}$

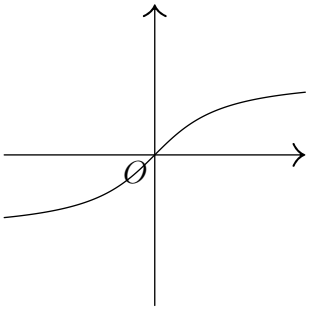
C. $\dfrac{4\pi}{3\sqrt{3}}$

D. $2\pi - \dfrac{\pi}{\sqrt{3}}$

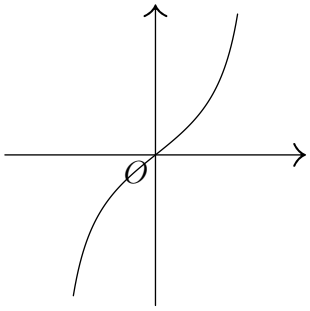
8 The complex number z satisfies $|z|\,\mathrm{Re}(z) = \mathrm{Im}(z)$.

Which of the following Argand diagrams could represent the locus of z?

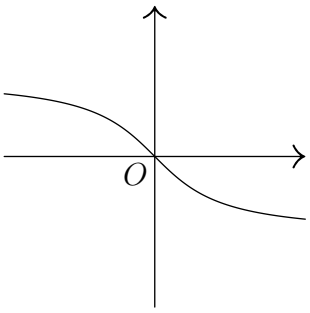
A.



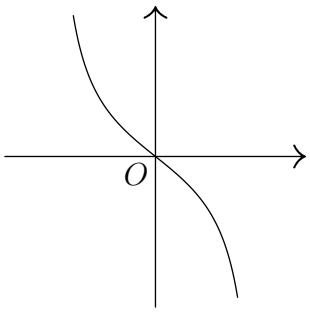
C.



B.



D.



9 Let z and w be non-zero complex numbers.

The point A is represented by the complex number z.

The point B is represented by the complex number $w + iz$.

Given that $OA \perp OB$, where O is the origin of the Argand plane, which of the following could NOT equal $|w + iz|$?

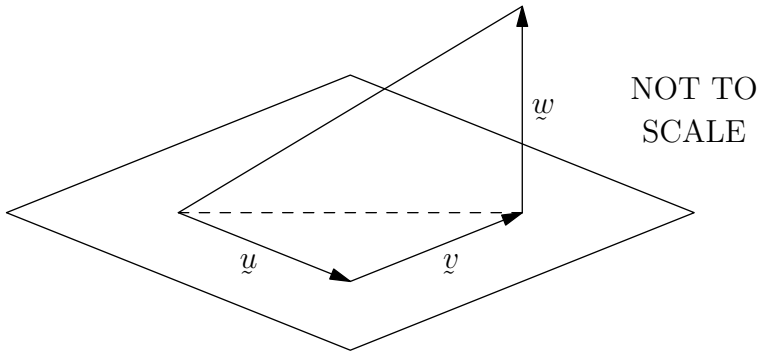
A. $|z| + |w|$

B. $|z| - |w|$

C. $|w| - |z|$

D. $|z + w|$

10 The non-zero vector $\underset{\sim}{u}$ and $\underset{\sim}{v}$ lie in a plane, and the direction of the vector $\underset{\sim}{w}$ is perpendicular to every line drawn in this plane. The vectors satisfy $\underset{\sim}{u} \cdot (\underset{\sim}{v} + \underset{\sim}{w}) = 0$.



Let α be the angle between $\underset{\sim}{u} + \underset{\sim}{v} + \underset{\sim}{w}$ and $\underset{\sim}{u} + \underset{\sim}{v}$, let β be the angle between $\underset{\sim}{u} + \underset{\sim}{v} + \underset{\sim}{w}$ and $\underset{\sim}{u}$, and let γ be the angle between $\underset{\sim}{v} + \underset{\sim}{w}$ and $\underset{\sim}{v}$.

Which of the following is true?

A. $\text{cosec}^2\,\alpha < \text{cosec}^2\,\beta + \text{cosec}^2\,\gamma$

B. $\cot^2\alpha > \cot^2\beta + \cot^2\gamma$

C. $\sin^2\alpha > \sin^2\beta + \sin^2\gamma$

D. $\cos^2\alpha > \cos^2\beta + \cos^2\gamma$

Section II

90 marks
Attempt Questions 11–16
Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the Question 11 Writing Booklet

(a) Let $z = (3 + 2i)(2 + iw)$, where $w = 2 + 3i$. **2**

Express $\bar{z}$ in the form $x + iy$, where x and y are real.

(b) Find complex numbers z and w such that $\dfrac{z}{w} = 2 + i$ and $zw = 2 - i$. **2**

(c) Two lines are given by

$$\mathbf{r}_1 = \begin{pmatrix} 1 \\ 8 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ k \\ 2 \end{pmatrix} \text{ and}$$

$$\mathbf{r}_2 = \begin{pmatrix} k+5 \\ 2 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} k \\ 9 \\ 6 \end{pmatrix},$$

where k is a real number.

(i) Given the lines intersect, by equating the x and z components, show that **2**

$$k\mu - 3\mu + k + 4 = 0.$$

(ii) Hence, or otherwise, find the value(s) of k for which the lines intersect. **2**

Question 11 continues on page 8

(d) Two particles P and Q move on the x axis so that at time t their positions are $x_P = \cos 2t$ and $x_Q = \sin t$.

(i) Find all times t, for $0 \leq t \leq 2\pi$, at which P and Q are at the same position. **2**

(ii) At one of these times, P and Q meet but do NOT cross each other. Find this time, justifying your answer. **1**

(e) Evaluate $\displaystyle\int_1^{25} \frac{\ln x}{(26-x)^2}\,\mathrm{d}x$. **4**

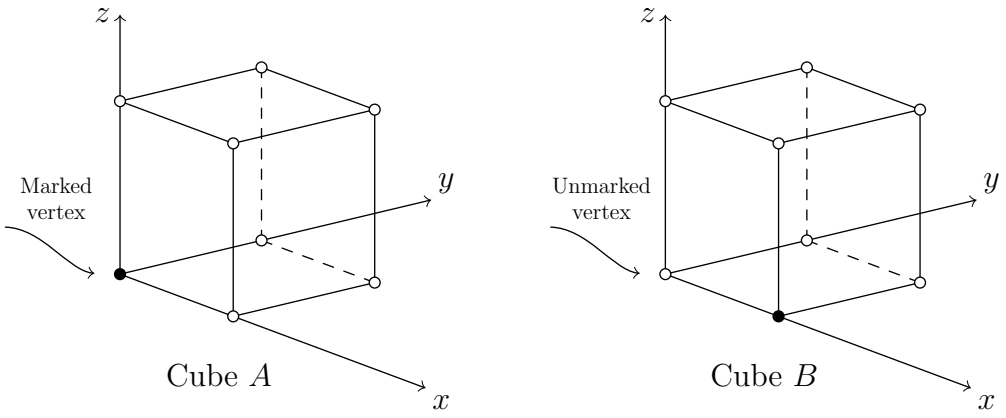
End of Question 11

Question 12 (15 marks) Use the Question 12 Writing Booklet

(a) Two identical cubes, A and B each have 8 vertices (x, y, z), where each of x, y and z is 0 or 1. Vertex $(0, 0, 0)$ is marked on cube A and vertex $(1, 0, 0)$ is marked on cube B. **2**

Exactly 4 vertices are to be marked on each cube. On the diagram on page 67 of the writing booklet, mark the three additional vertices on each cube, so that the marked vertices on the 2 cubes have identical projections on each of the xy, yz, and zx-planes.

Cube A and cube B must NOT have exactly the same 4 vertices marked.



(b) Prove by contradiction that for positive integers n, **3**

$$\sqrt{n+1} + \sqrt{n+2} > \sqrt{n} + \sqrt{n+3}.$$

(c) A particle of mass m is released from rest and falls vertically. It experiences air resistance of magnitude mkv, where v is speed, and k is a positive constant. The acceleration due to gravity is g. Let $y(t)$ be the distance fallen after t seconds.

(i) By considering definite integrals, show that $\dot{y}(T) < gT$, for $T > 0$. **1**

(ii) Using part (i), show that $\ddot{y}(T) > (1 - kT)g$, for $T > 0$. **1**

(iii) Hence show that $\dfrac{1}{2}gT^2\left(1 - \dfrac{kT}{3}\right) < y(T) < \dfrac{1}{2}gT^2$, for $T > 0$. **2**

Question 12 continues on page 10

(d) A sphere cuts the x-axis at the points $(1, 0, 0)$ and $(6, 0, 0)$ and touches the z-axis at exactly one point. Find the possible point(s) at which the sphere touches the z-axis. **3**

(e) A particle is projected form the origin with speed V at angle α above the horizontal. The resistive force on the particle is proportional to its velocity. **3**

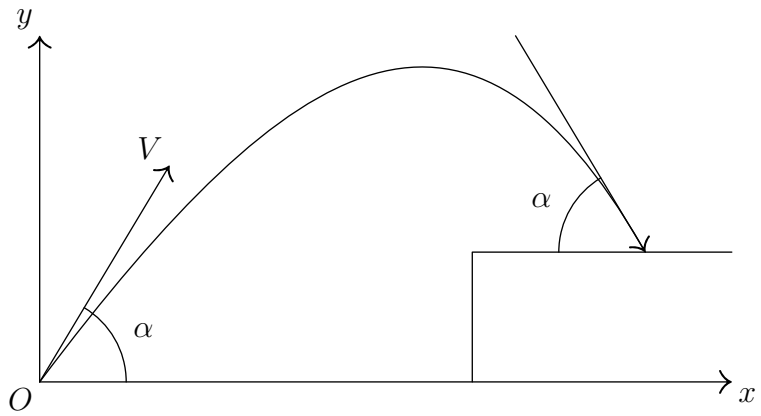
The position of the particle t seconds after projection is (x, y) and the velocity of the particle at time t seconds after projection is $\underset{\sim}{v}(t) = \begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix}$.

Taking the upwards vertical direction and the horizontal direction of projection to be positive, the velocity of the particle at time t is given by

$$\underset{\sim}{v}(t) = \begin{pmatrix} Ve^{-kt}\cos\alpha \\ \left(V\sin\alpha + \dfrac{g}{k}\right)e^{-kt} - \dfrac{g}{k} \end{pmatrix} \quad \text{(Do NOT prove this.)}$$

where k is a positive constant and g is acceleration due to gravity.

The particle lands on a horizontal platform making an angle α with the platform, as shown in the diagram below.



Show that the horizontal distance when the particle lands is given by

$$\frac{V^2\sin 2\alpha}{g + 2kV\sin\alpha}.$$

End of Question 12

Question 13 (15 marks) Use the Question 13 Writing Booklet

(a) A particle of mass 1kg is moving in simple harmonic motion along the x-axis. Initially, the particle is at the origin, and the restoring force acting on it has magnitude 9 newtons.

The particle next returns to the origin $\dfrac{\pi}{3}$ seconds later, and returns again $\dfrac{2\pi}{3}$ seconds after that.

(i) Explain why the period of this motion is π seconds. **1**

(ii) Find the distance between the origin and the centre of motion. **2**

(iii) Find the initial speed of the particle. **2**

(b) (i) Let z be a complex number with modulus 1. Show that **2**

$$|z+1| = \sqrt{2 + 2\,\mathrm{Re}(z)}.$$

(ii) Show that **3**

$$1 + x^5 = (1+x)\left(x + e^{\frac{2i\pi}{5}}\right)\left(x + e^{\frac{4i\pi}{5}}\right)\left(x + e^{\frac{6i\pi}{5}}\right)\left(x + e^{\frac{8i\pi}{5}}\right).$$

(iii) Hence, evaluate **2**

$$\left(1 + \cos\frac{2\pi}{5}\right)\left(1 + \cos\frac{4\pi}{5}\right)\left(1 + \cos\frac{6\pi}{5}\right)\left(1 + \cos\frac{8\pi}{5}\right).$$

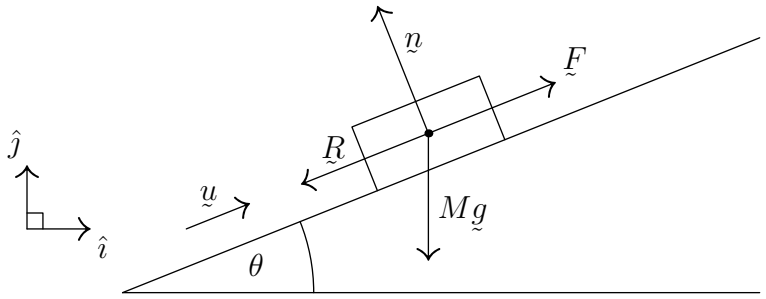
(c) Let f be a continuous function satisfying $f(1+x) = \dfrac{1 - f(x)}{1 + f(x)}$ for all x. Prove that **3**

$$\int_0^2 f(x)\,\mathrm{d}x = \frac{1}{2}\int_0^4 f(x+2)\,\mathrm{d}x\,.$$

Question 14 (15 marks) Use the Question 14 Writing Booklet

(a) A car of mass M drives up a straight ramp of length L metres to access subsequent storeys of a parking lot. the vector $\underset{\sim}{u} = \cos\theta\hat{\imath} + \sin\theta\hat{\jmath}$ is parallel to the ramp and points directly up the incline.

The position of the car is given by $\underset{\sim}{r} = x\underset{\sim}{u}$. When moving, the car experiences air resistance $\underset{\sim}{R} = -Mkv^2\underset{\sim}{u}$, where v is the speed of the car, and k is a positive constant. The normal reaction force is $\underset{\sim}{n} = Mg\cos\theta\underset{\sim}{\hat{n}}$, where $\underset{\sim}{\hat{n}} = -\sin\theta\hat{\imath} + \cos\theta\hat{\jmath}$. The force due to gravity is $M\underset{\sim}{g}$ and has magnitude Mg.



The car is driven from rest at $x = 0$ by its engine, which provides a constant force $\underset{\sim}{F} = 2Mg\sin\theta\underset{\sim}{u}$ while the accelerator is engaged.

At $x = d$, the driver disengages the accelerator, and the car continues moving up the ramp, under ONLY the influences of gravity, the reaction force, and air resistance, with the exact speed V required to come to rest at the top of the ramp.

(i) By considering dot products with $\underset{\sim}{u}$, show that **2**

$$\ddot{x} = g\sin\theta - kv^2,$$

for $0 < x < d$.

(ii) Show that $V^2 = \dfrac{g\sin\theta}{k}\left(e^{2k(L-d)} - 1\right)$. **2**

(iii) Find d in terms of k and L. **3**

Question 14 continues on page 13

Question 14 (continued)

(b) Suppose $\sqrt{2}$ is rational. It can be shown that $\sqrt{2} > 1$ (do NOT prove this), so there are smallest integers $a > 0$ and $b > 0$ for which $\sqrt{2} = 1 + \dfrac{a}{b}$.

(i) Show that $\dfrac{b}{a} = 2 + \dfrac{a}{b}$. **2**

(ii) Show that $0 < b - 2a < a$. **2**

(iii) Deduce that $\sqrt{2}$ is irrational. **1**

(c) The non-real numbers a and b satisfy **3**

$$|a + ib| = \sqrt{a^2 + b^2}.$$

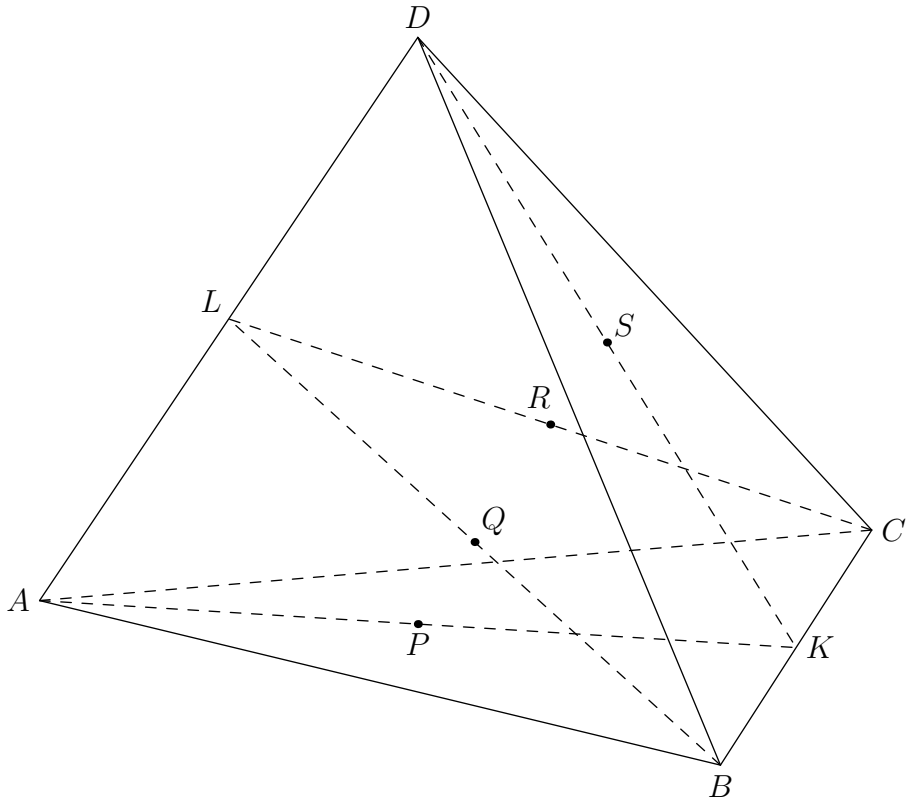
Prove that $b = ia$.

End of Question 14

Please turn over

Question 15 (15 marks) Use the Question 15 Writing Booklet

(a) In tetrahedron $ABCD$, K is the midpoint of BC and L is the midpoint of AD, as shown in the diagram.



The point P is the midpoint of AK, the point Q is the midpoint of BL, the point R is the midpoint of CL, and the point S is the midpoint of DK.

(i) Show that $\overrightarrow{AP} = \frac{1}{4}\left(\overrightarrow{AB} + \overrightarrow{AC}\right)$. **1**

(ii) It can be shown that **3**

$$\overrightarrow{BQ} = \frac{1}{4}\left(\overrightarrow{BA} + \overrightarrow{BD}\right)$$
$$\overrightarrow{CR} = \frac{1}{4}\left(\overrightarrow{CA} + \overrightarrow{CD}\right) \qquad \text{(Do NOT prove these)}$$
$$\text{and} \quad \overrightarrow{DS} = \frac{1}{4}\left(\overrightarrow{DB} + \overrightarrow{DC}\right).$$

Prove that P, Q, R and S are the vertices of a parallelogram.

(iii) Prove that the points P, Q, R and S are vertices of a rhombus, if and only if $AD \perp BC$. **2**

Question 15 continues on page 15

(b) (i) Let θ be a real number. Consider the numbers **2**

$$1, e^{i\theta}, e^{2i\theta}, \cdots, e^{6i\theta}.$$

By dividing the unit circle into equal arcs, or otherwise, show that two of these numbers are less than 1 unit apart on the complex plane.

(ii) Hence prove that, for any real number θ, there exists some integer k with $1 \leq k \leq 6$, such that $\cos(k\theta) > \dfrac{1}{2}$. **3**

(c) For integers $n \geq 1$, define $S_n = \displaystyle\sum_{r=1}^{n} \frac{1}{2^r - 1}$.

(i) Show that if $A > \dfrac{8}{7}$, then $\dfrac{A}{2^{k+1}} > S_{k+1} - S_k$, for all integers $k \geq 2$. **1**

(ii) By using mathematical induction and considering part (i) with a suitable value of A, show that $S_n < \dfrac{23}{14}$ for all integers $n \geq 1$. **3**

End of Question 15

Please turn over

Question 16 (15 marks) Use the Question 16 Writing Booklet

(a) (i) Show that **2**

$$\ln(x+1) > x \ln\left(\frac{x}{x+1}\right)$$

for $x > 0$.

(ii) By considering $f(x) = \dfrac{\ln(x+1)}{\ln x}$, or otherwise, show that for $x > 1$, **3**

$$\log_x(x+1) > \log_{x+1}(x+2).$$

(b) Define $I_n = \displaystyle\int_0^1 \sin(x^n)\,\mathrm{d}x$ for integers $n = 2, 3, 4, \ldots$.

(i) Using the substitution $t = x^n$ and integration by parts, show that **3**

$$\sin(1) - I_n = \frac{n}{n+1} - 2\int_0^1 \sqrt[n]{t}\,\sin^2\left(\frac{t}{2}\right)\mathrm{d}t\,.$$

(ii) Show that **2**

$$\sin(1) - \frac{n}{n+1} \leq I_n \leq \sin(1).$$

(iii) By considering part (i), or otherwise, show that **3**

$$\sin(1) - I_n > \frac{11}{21}.$$

(iv) Deduce that $\dfrac{1}{\sin(1) - I_n}$ can NOT be an integer. **2**

End of paper

BLANK PAGE

BLANK PAGE

BLANK PAGE



© 2077 W² Education Standards Authority